

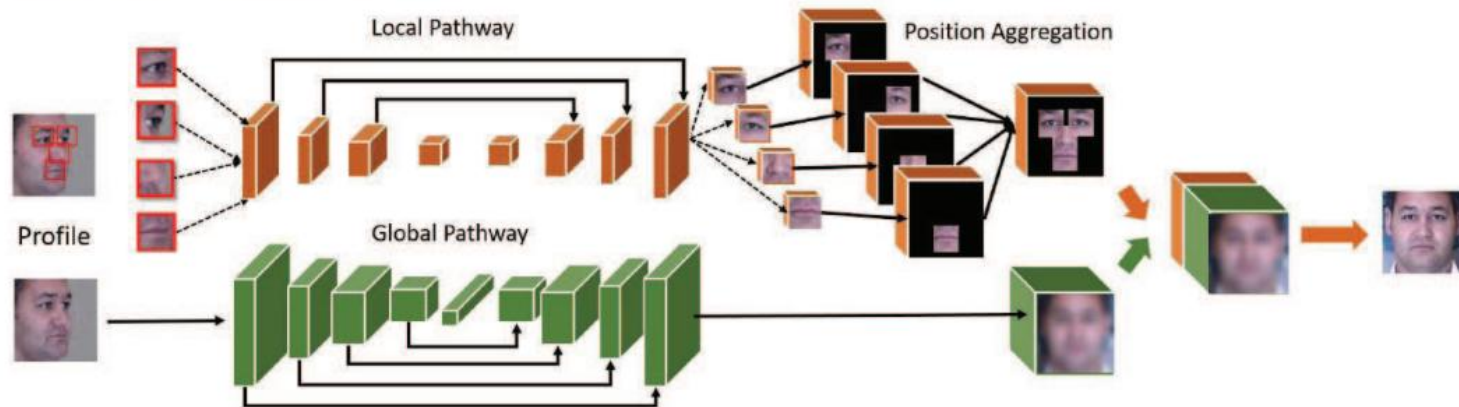
Deep Face Recognition

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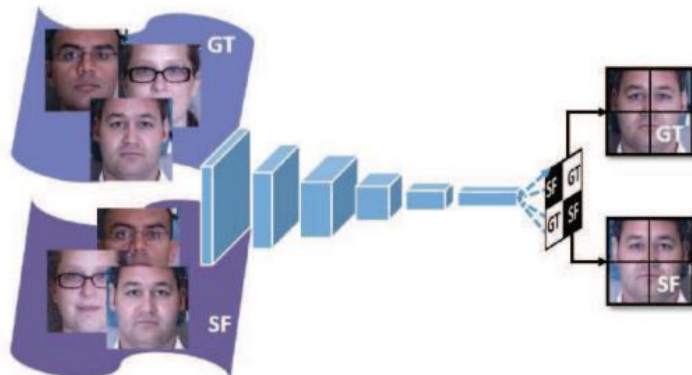
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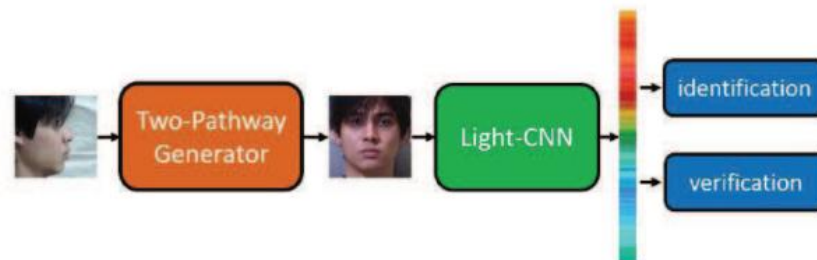
Two-pathway Generator Network



Discriminator Network



Recognition via Generation

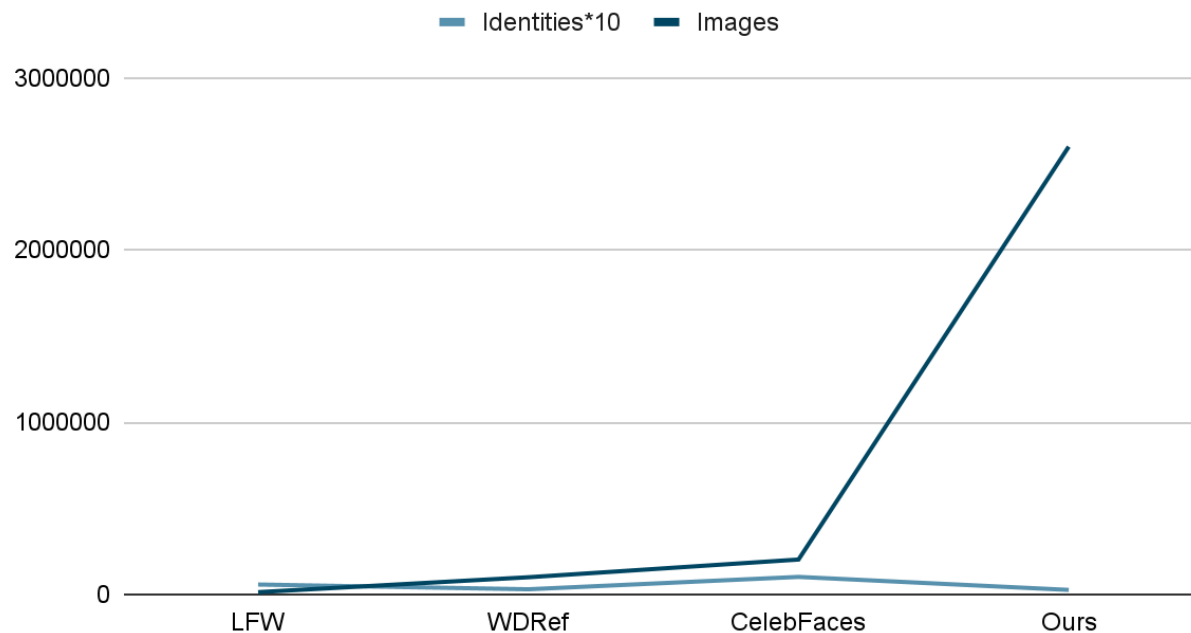


Introduction

Goals:

- To propose a procedure to create a large face dataset
- To investigate various CNN architectures for face identification and verification

Dataset comparisons



	Identities	Images
Ours	2622	2.6M
Google	8M	200M
FaceBook	4030	4.4M

Largest collection of face images outside industrial datasets by Google and Facebook

Related Work



Example images from our dataset for six identities

Dataset Collection

Stage 1. Bootstrapping and filtering a list of candidate identity names

Stage 2. Collecting more images for each identity

Stage 3. Improving purity with an automatic filter

Stage 4. Near duplicate removal

Stage 5. Final manual filtering

Stage	Aim	Type	# of persons	# of images per person	total # images	Annotation effort	100%-EER
1	Candidate list generation	A	5,000	200	1,000,000	–	–
2	Image set expansion	M	2,622	2,000	5,244,000	4 days	–
3	Rank image sets	A	2,622	1,000	2,622,000	–	96.90
4	Near dup. removal	A	2,622	623	1,635,159	–	–
5	Final manual filtering	M	2,622	375	982,803	10 days	92.83

Dataset statistics after each stage of processing

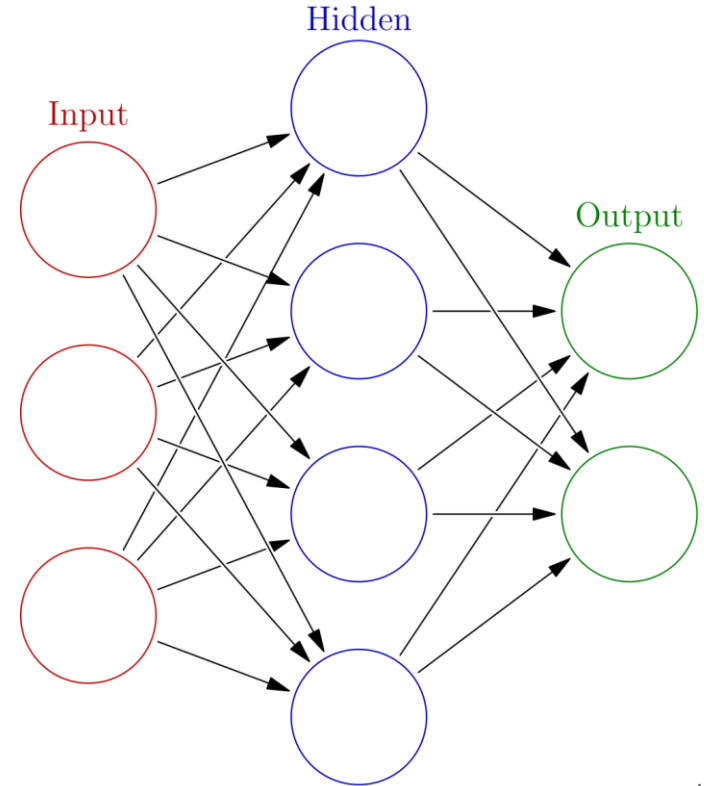
Main Methodologies

- Covolutional Neural Network
- Bootstrapping
- Triplet Loss

What are Artifitial Neural Networks?

Artificial Neural Network

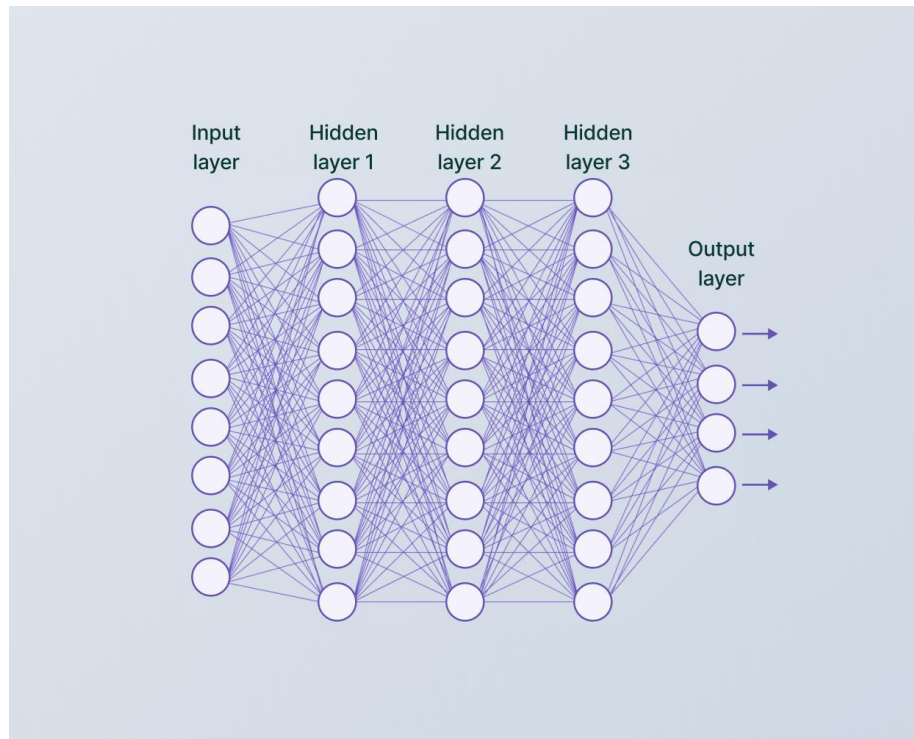
- ANNs are inspired by biological neural networks
- Basic Structure: Input, Hidden, and Output layers
- Weights between layers to adjust Inputs



What differentiates ANN and CNN?

Convolutional Neural Network

- CNNs have hidden layers called “Convolutional Layers”
- a Convolutional Layer receives input and then transforms the input in some way and outputs the transformed input to the next layer



Convolutional Neural Network

- Each convolutional layer is a filter or a set of filters
- Each filter “convolves” with the image and creates an activation map
- The output of the Convolutional Layer is generated by stacking the activation maps of the filters along the path

7	2	3	3	8
4	5	3	8	4
3	3	2	8	4
2	8	7	2	7
5	4	4	5	4

 $*$

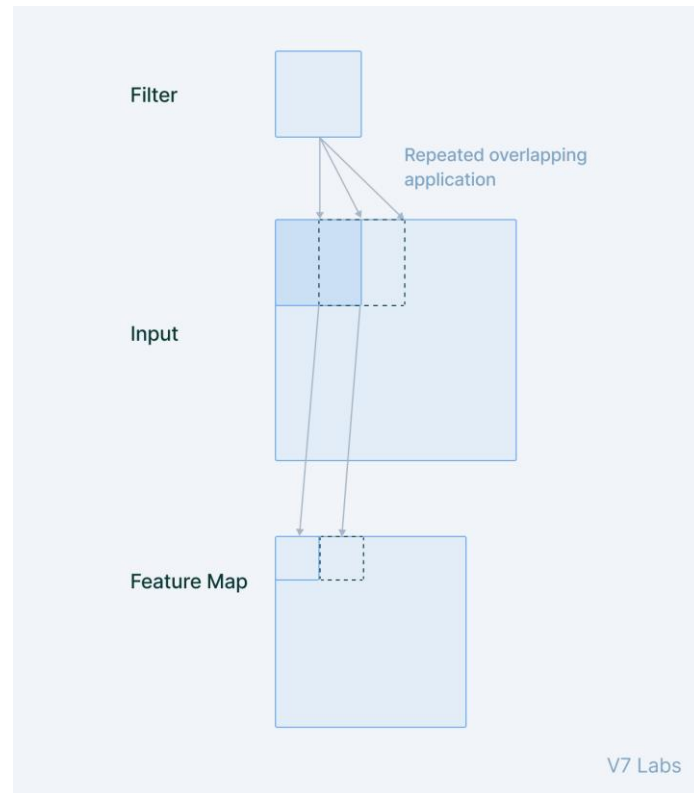
1	0	-1
1	0	-1
1	0	-1

 $=$

6		

$7 \times 1 + 4 \times 4 + 3 \times 1 + 2 \times 0 + 5 \times 0 + 3 \times 0 + 3 \times -1 + 3 \times -1 + 2 \times -1 = 6$

V7 Labs



V7 Labs

Example of CNN Application

- Number recognition
- Edge detection



Conclusion: Convolutional Neural Network

- ANN with specialization that can pick out or detect patterns and make sense of it
- Popular for analyzing images because of it's architecture

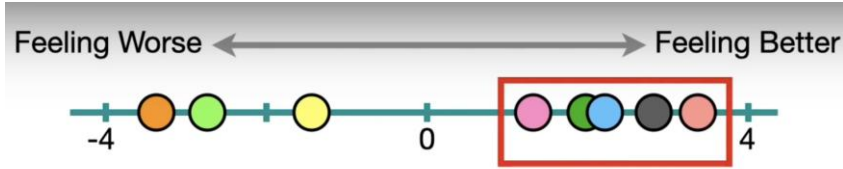
Bootstrapping

- A resampling technique
- Used to estimate statistics on a population by sampling a dataset with replacement

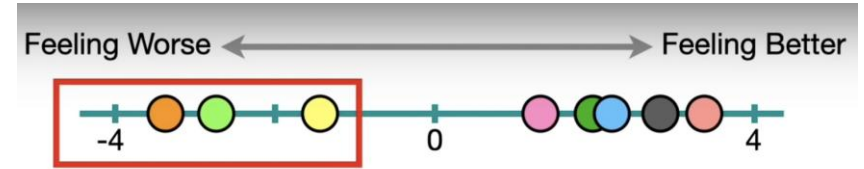


Why Bootstrapping?

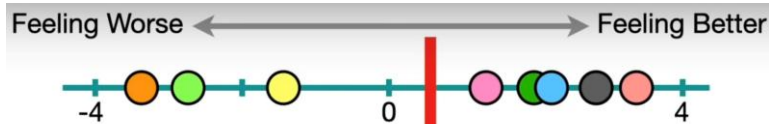
Bootstrapping



For **5** of those people, the drug appeared to help them feel better...



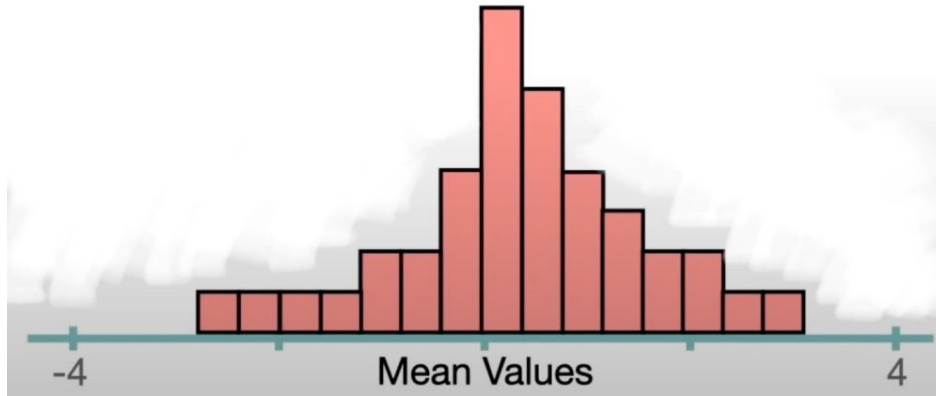
...but for **3** people, the drug appeared to make them feel worse.



If we calculate the mean of the response to the drug we get **0.5**.

- After replicating this experiment on different people, we get a gaussian distribution of the mean value

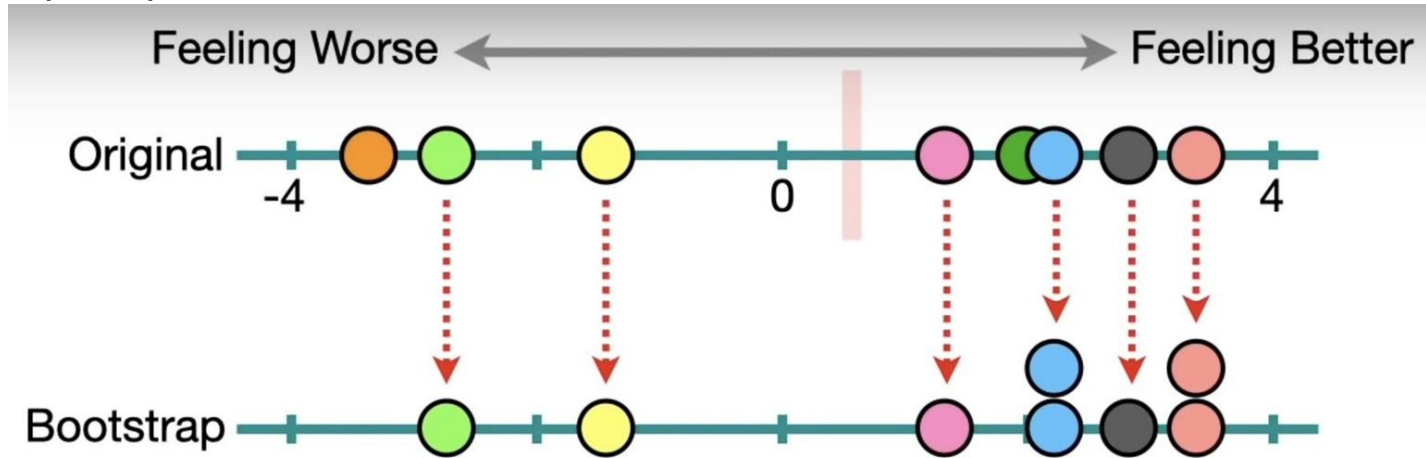
Bootstrapping



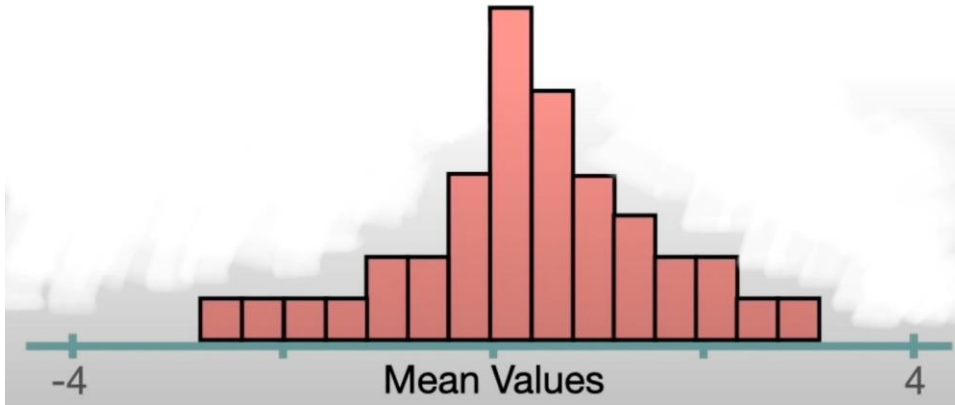
- However, replicating the same experiment is possible but too resource consuming
- This is where Bootstrapping comes into play

Bootstrapping

- 1) Make a Bootstrapped dataset
 - 2) Calculate something
 - 3) Keep in track with the calculation
 - 4) Eat, Sleep, Repeat!
- Selection is random
 - Duplicates are allowed
 - So-called Sampling with Replacement



Bootstrapping



- Bootstrapping “enlarges” the dataset
- Reduction on time and resources

What is a Triplet Loss?

$$E(W') = \sum_{(a,p,n) \in T} \max\{0, \alpha - \|\mathbf{x}_a - \mathbf{x}_n\|_2^2 + \|\mathbf{x}_a - \mathbf{x}_p\|_2^2\}, \quad \mathbf{x}_i = W' \frac{\phi(\ell_i)}{\|\phi(\ell_i)\|_2}.$$

$$E(W') = \sum_{(a,p,n) \in T} \max\{0, \alpha - \|\mathbf{x}_a - \mathbf{x}_n\|_2^2 + \|\mathbf{x}_a - \mathbf{x}_p\|_2^2\}, \quad \mathbf{x}_i = W' \frac{\phi(\ell_i)}{\|\phi(\ell_i)\|_2}.$$

Don't have a clue what it means?

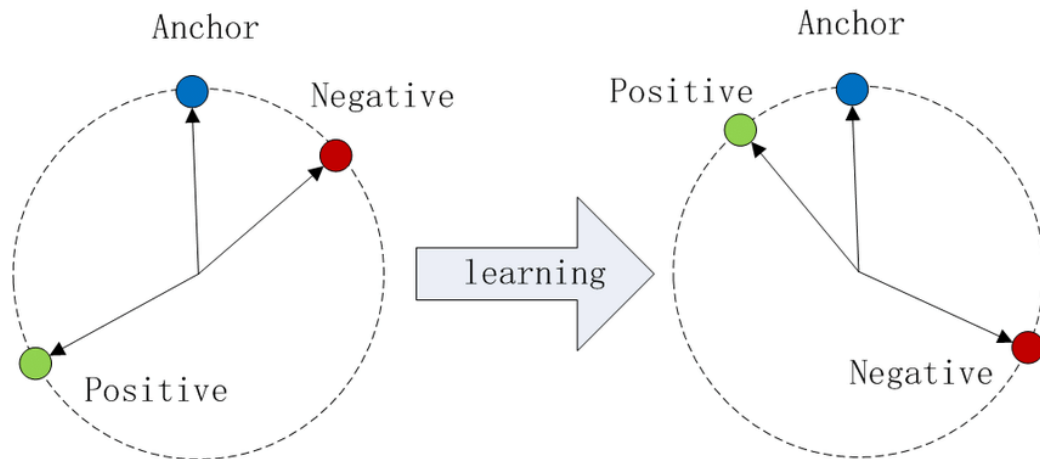
Don't worry, me neither.

Triplet Loss

- A type of loss function for machine learning algorithm

Core idea: Compare

- Reference Input (Anchor)
- Matching Input (Positive)
- Non-Matching Input (Negative)



Triplet Loss

- Look at 3 images at once
- Anchor and positive same person
- Anchor and negative different person
- Compare the distances
- Use gradient descent to have a backpropagating effect on the neural network

Anchor



Positive



Negative



Conclusion - Chapter: Methodology

- CNN is popularly used for pattern recognition
- Bootstrapping is used for the dataset issues
- Triplet Loss is used for optimizing the results

Datasets and evaluation protocols

- Evaluation is performed on existing benchmark datasets
- The first dataset is Labeled **Faces in the Wild dataset** (LFW) - They follow the standard evaluation protocol defined for the “unrestricted setting” using data external to LFW for training.
- The second dataset is **YouTube Faces** (YTF) - They follow the standard evaluation protocol defined for the “unrestricted setting”, and report EER.

Experiments and Results

- They first test several variations of the proposed models and training data using the LFW dataset to test performance.
- Implementation is based on the MATLAB toolbox MatConvNet combined with the NVIDIA CuDNN libraries to accelerate training.
- Experiments done on NVIDIA Titan Black GPUs

No.	Config	Data	Train Align.	Test Align.	Embedding	100% - EER
1	A	C	No	No	No	92.83
2	A	F	No	No	No	95.80
3	A	F	No	Yes	No	96.70
4	B	F	No	Yes	No	97.27
5	B	F	Yes	Yes	No	96.17
6	D	F	No	Yes	No	96.73
7	B	F	No	Yes	Yes	99.13

Component analysis

- This evaluates the effect of different components of the system when training a network for face verification.
1. Dataset curation
 2. Alignment
 3. Architecture
 4. Triplet-loss embedding

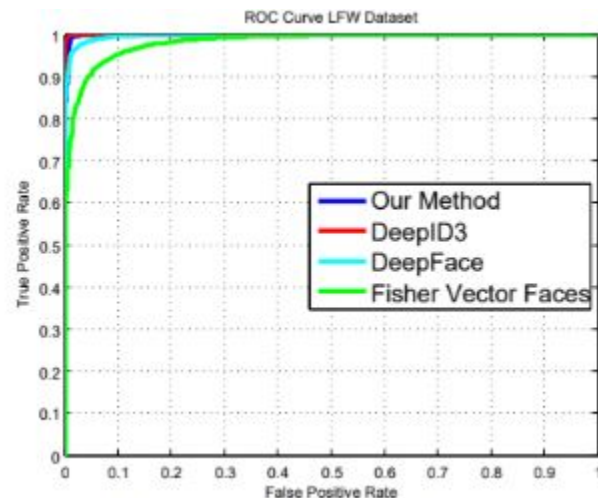
No.	Config	Data	Train Align.	Test Align.	Embedding	100% - EER
1	A	C	No	No	No	92.83
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4	B	F	No	Yes	No	97.27
5	B	F	Yes	Yes	No	96.17
6	D	F	No	Yes	No	96.73
7	B	F	No	Yes	Yes	99.13

Comparison with the state-of-the-art

LFW dataset

- It can be observed that they achieve comparable results to the state of the art using much less data and a much simpler network architecture.

No.	Method	Images	Networks	Acc.
1	Fisher Vector Faces []	-	-	93.10
2	DeepFace []	4M	3	97.35
3	Fusion []	500M	5	98.37
4	DeepID-2,3		200	99.47
5	FaceNet []	200M	1	98.87
6	FaceNet [] + Alignment	200M	1	99.63
7	Ours	2.6M	1	98.95



YTF dataset

- The state of the art performance using our triplet loss embedding method was achieved.

No.	Method	Images	Networks	100%- EER	Acc.
1	Video Fisher Vector Faces [15]	-	-	87.7	83.8
2	DeepFace [29]	4M	1	91.4	91.4
3	DeepID-2,2+,3		200	-	93.2
4	FaceNet [17] + Alignment	200M	1	-	95.1
5	Ours ($K = 100$)	2.6M	1	92.8	91.6
6	Ours ($K = 100$) + Embedding learning	2.6M	1	97.4	97.3

Conclusions

Two contributions:

- A procedure has been decided that is able to assemble a large scale dataset, with small label noise, whilst minimising the amount of manual annotation involved.
- A deep CNN was shown, without any embellishments but with appropriate training, can achieve results comparable to the state of the art.
- Also suitable for other object classes as well as fine grained tasks.

Thank You !